

Activity – 15

Writing a menu driven c program to perform one of the operations based upon the choice of the user

1. Check whether all digits are even.
2. Check whether all digits are prime.

Program

```

int main(){
    int num, choice;
    printf("Please enter the number\n");
    scanf("%d", &num);

    do{
        printf("\n-----Menu-----\n");
        printf("\nfirst choice is for checking if all are even digits\n");
        printf("\nsecond choice is for checking if all digits are prime.\n");
        scanf("%d", &choice);

        switch(choice){
            case 1: {
                int temp = num, digit;
                int all_even = 1;
                while(temp > 0){
                    digit = temp % 10;
                    if(digit % 2 != 0){
                        all_even = 0;
                        break;
                    }
                    temp /= 10;
                }
                if(all_even) printf("All digits are even\n");
                else printf("All digits are not even.\n");
                break;
            }
            case 2: {
                int temp = num, digit;
                int all_prime = 1;
                while(temp > 0){
                    digit = temp % 10;
                    if(!digit == 2 || digit == 3 || digit == 5 || digit == 7){
                        all_prime = 0;
                        break;
                    }
                    temp /= 10;
                }
                if(all_prime) printf("All digits are prime.\n");
                else printf("All digits are not prime.\n");
                break;
            }
            case 3: {
                printf("Exiting the program.\n");
                break;
            }
            default:
                printf("Invalid choice.\n");
        }
    }while(choice != 3);
    return 0;
}

```

Compilation and execution

```

(kali@kali)-[~]
$ vi newd6.c

(kali@kali)-[~]
$ gcc newd6.c -o newd6

```

Algorithm

1.START

2.Input of user for number.

3.do-while loop use

4.give choice to user through input by switch case.

5. choice1 = to check if all digits are even.

6. choice2 = to check if all digits are prime.

7. choice3 = to exit

8. default for saying invalid choice.

8.STOP

TEST CASE

Inpu t	Expected	Actual	RESU LT
1234 5	(False)(Fal se)	(False), (False)	PASS
1890	(False)(Fal se)	(False),(Fal se)	PASS
1763 4	False, False	False, False	PASS